



Short-term predictions of water levels at a Dutch Storm Surge Barrier

DATA ANALYSIS ASSIGNMENT, REGRESSION MODELS AND APPLICATIONS

Jasper Arends, based on the lecture notes by Prof. Alessandro Di Bucchianico

Water level in the Netherlands

NAP = Normaal Amsterdams Peil
(reference point for sea water level)

Flood-prone area

- Within dyke rings
 - Below NAP: 26%
 - Above NAP: 29%
- Areas outside dyke rings: 3%
- Areas threatened by flooding along the river Meuse: 1% *)



*) Flood-prone area along the river Meuse, within the 1/250-contour.

North Sea flood of 1953 (watersnoodsramp)

- Flood caused by a storm surge + spring tide
- Highest water level in 100 years (+4.55m NAP)
- 1836 people died in the Netherlands



Dikes at Alblasserwaard break

Delta Works

One of the **Seven Wonders of the Modern World**
Combination of dams and **flood barriers**



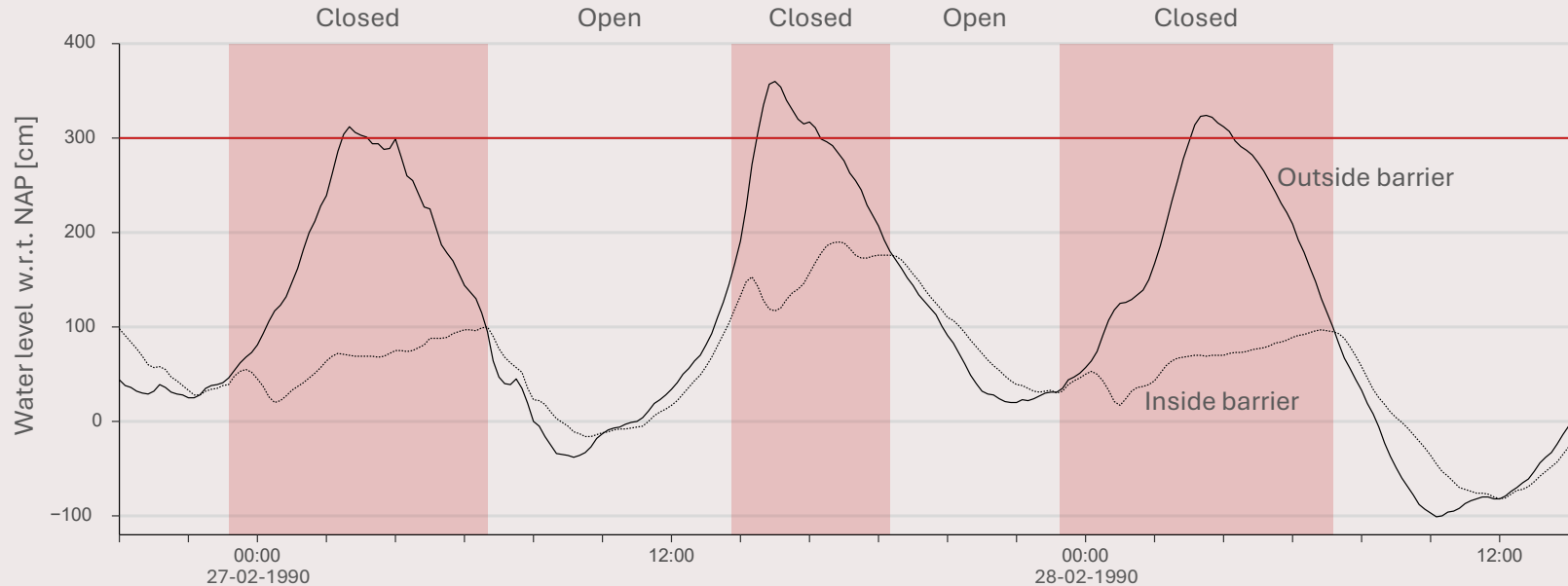
Eastern Scheldt Storm Surge Barrier

Sluice gates letting water in and out
Blocking water is harmful to ecosystems

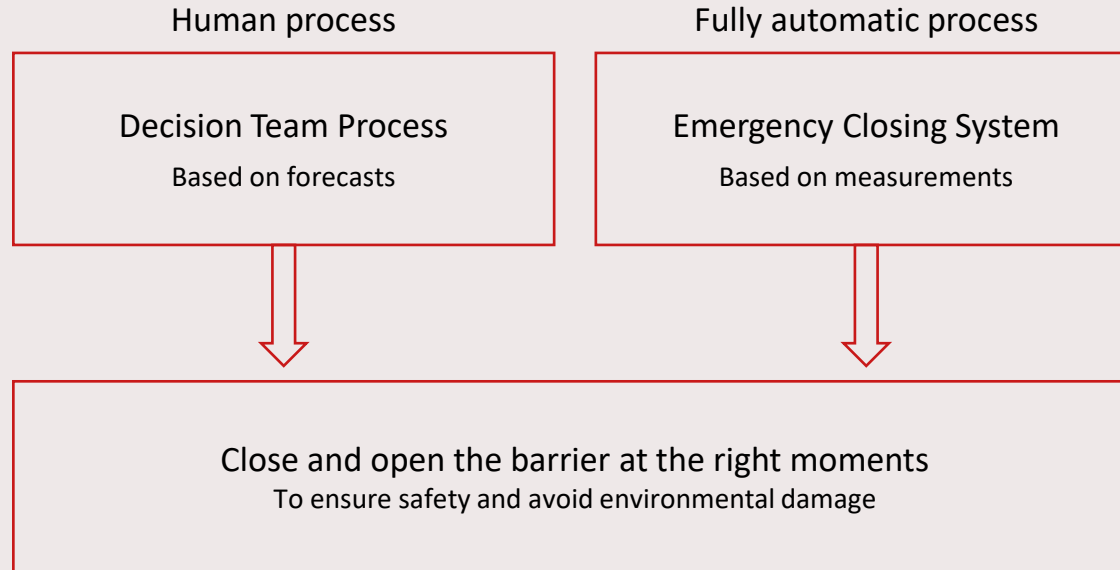


Closing of the Roompot storm surge barrier

Average water level within 10 minute intervals (waterinfo.rws.nl)



Closing of the Roompot storm surge barrier



Closing of the Roompot storm surge barrier

Forecasts are made using the Dutch Continental Shelf Model

Forecast water level $> +2.75\text{m NAP}$
→ call for “decision team”

Forecast water level $> +3\text{m NAP}$
→ close using 1-2-1 strategy

Automatic closing if measurements $> +3\text{m NAP}$

More details: www.tl.iplo.nl/@237761/watersysteemtype-oosterschelde (Dutch)



Modelling assignment

Make short-term predictions of the water level at the storm surge barrier RPBU

Part I: Predictions using data only from RPBU

- Data analysis and physical modelling of RPBU harbour suggest there are multiple oscillations at play

Part II: External data can be used to improve predictions

- Waves travel from VR to OS11 and RPBU

OS11	Oosterschelde 11
RPBU	Oosterschelde Roompotsluis Buiten
VR	Vlakte van de Raan



The report

- Report layout
 - Introduction
 - Main text
 - Conclusion
- Provide R scripts in the appendix

Deadline: 29/06/2026, 23:30

For this week

- Split into groups of 3 students
 - Sign up on Canvas, **deadline: 01/05/2026, 23:59**
- Perform exploratory data analysis
- Students from 2MBS50 who have successfully completed the assignment in previous years automatically keep the same grade
 - Inform us by 01/05/2026 via e-mail